Hints: Linear algebra – 1. Determinants and invertibility

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Regard $\det A$ as a polynomial F in the n^2 entries a_{ij}. By the chain rule, $\frac{d}{dt}\det A(t) = \sum_{i,j} \frac{\partial F}{\partial a_{ij}}\, a'_{ij}(t)$, so it suffices to compute each partial derivative.

Hint 2. Expand along row i: $\det A = \sum_k a_{ik}(-1)^{i+k}M_{ik}$, and the cofactors $(-1)^{i+k}M_{ik}$ do not involve any entry of row i. Hence $\partial F/\partial a_{ij} = (-1)^{i+j}M_{ij} = (\operatorname{adj} A)_{ji}$, and $\sum_{i,j}(\operatorname{adj} A)_{ji}a'_{ij} = \operatorname{tr}(\operatorname{adj} A \cdot A')$.

Hint 3. For $\det e^A$: put $f(t) = \det e^{tA}$. Jacobi gives $f'(t) = \operatorname{tr}(\operatorname{adj}(e^{tA})e^{tA}A) = f(t)\operatorname{tr} A$ (since e^{tA} is invertible, $\operatorname{adj}(e^{tA}) = f(t)e^{-tA}$), and $f(0) = 1$. For $\operatorname{adj}(\operatorname{adj} A)$: first treat invertible A, where $B = \operatorname{adj} A = (\det A)A^{-1}$ and $\det B = (\det A)^{n-1}$; plug into $\operatorname{adj} B \cdot B = (\det B)E$. Then extend to all A because both sides are polynomials in the entries and invertible matrices are dense (or: polynomial identity holding on a Zariski-dense set).

Problem 2

Hint 1. The matrix is "staircase"-shaped: column j equals column $j-1$ except in rows $i \geqslant j$. Subtract column $j-1$ from column j, for $j = n, n-1, \dots, 2$.

Hint 2. After these operations the matrix is lower triangular with diagonal entries 1 and $\frac{1}{k} - \frac{1}{k-1} = -\frac{1}{k(k-1)}$ for $k = 2, \dots, n$. Multiply them: the answer is $\det A = \dfrac{(-1)^{n-1}}{(n-1)!\,n!}$.

Problem 3

Hint 1. The matrix is $D + uu^T$ with $D = \operatorname{diag}(2, 3, \dots, n)$ and $u = (1, \dots, 1)^T$. Use the rank-one update formula (Theorem 2).

Hint 2. $\det D = n!$ and $u^T D^{-1} u = \frac{1}{2} + \dots + \frac{1}{n}$, so $D_n = n!\,(1 + \frac{1}{2} + \dots + \frac{1}{n})$. So $D_n/n! = H_n$, the harmonic number, which is unbounded: the answer is no.

Problem 4

Hint 1. $\cos((j+k)\theta) = \cos j\theta \cos k\theta - \sin j\theta \sin k\theta$, so A has rank at most 2: $A = UV^T$ with U, V of size $n \times 2$. Use $\det(E + UV^T) = \det(E_2 + V^T U)$.

Hint 2. Take U with columns $(\cos k\theta)_k, (\sin k\theta)_k$ and V with columns $(\cos k\theta)_k, (-\sin k\theta)_k$. Then $V^T U$ is the 2×2 matrix of sums $\sum \cos^2 k\theta$, $\sum \cos k\theta \sin k\theta$, $\sum \sin^2 k\theta$ (with signs).

Hint 3. With $\theta = 2\pi/n$ and $n \geqslant 3$ one has $\sum_{k=1}^{n} \cos 2k\theta = \sum_{k=1}^{n} \sin 2k\theta = 0$, so $\sum \cos^2 k\theta = \sum \sin^2 k\theta = n/2$ and $\sum \cos k\theta \sin k\theta = 0$. The answer is $1 - \dfrac{n^2}{4}$.

Problem 5

Hint 1. $\operatorname{adj} A$ is nilpotent, hence singular, hence by the rank facts about the adjugate its rank is 1 or 0.

Hint 2. A rank-one matrix is uv^T, and $(uv^T)^m = (v^Tu)^{m-1}uv^T$. So nilpotency forces $v^Tu = 0$, and then already $(uv^T)^2 = (v^Tu)uv^T = O$.

Problem 6

Hint 1. Look for a product that the two relations make vanish. Try computing $(A^2+B^2)(A-B)$.

Hint 2. $(A^2 + B^2)(A - B) = A^3 - A^2B + B^2A - B^3 = 0$. Since $A - B \neq 0$, $A^2 + B^2$ cannot be invertible. The answer is no.

Problem 7

Hint 1. By Theorem 3, the hypothesis says $\det(A + kB) = \pm 1$ for $k = 0, \dots, 4$. Consider $f(t) = \det(A + tB)$: what kind of function of t is it?

Hint 2. f is a polynomial of degree at most 2 with integer coefficients taking values in $\{1, -1\}$ at five points. By pigeonhole it takes the same value at three of them, so it is constant. Hence $\det(A + 5B) = \pm 1$.

Problem 8

Hint 1. It suffices to show the kernel is trivial (dimension count). Suppose $\varphi(A) = 0$ with $A \neq 0$. What does determinant-preservation give for $\det(X + A)$ for arbitrary X?

Hint 2. $\det(X + A) = \det(\varphi(X) + \varphi(A)) = \det \varphi(X) = \det X$ for all X. Now find an X that is singular but with $X + A$ invertible.

Hint 3. If $\operatorname{rank} A = r \geqslant 1$, write $A = P \operatorname{diag}(E_r, 0)Q$ with P, Q invertible and take $X = P \operatorname{diag}(0, E_{n-r})Q$. Then $\det X = 0$ but $X + A = PQ$ is invertible, a contradiction.